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Forecasting river sediment deposition through satellite image driven unsupervised machine learning techniques

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ABSTRACT

The hydrologic and riverine systems of the Earth's surface, subject to ever increasing instability, requires forecasting techniques now more than ever. Existing mathematical and machine learning models for forecasting river sediment deposition are driven by non-remotely sensed data and/or require ground truth data. The study aims to demonstrate that remote sensing and unsupervised machine learning techniques coupled with appropriate validation metric can be employed to quickly forecast regions that are subject to future river sediment deposition. The results indicate that the NIR and other bands outside the visible range are determinant features for forecasting depositional areas, a multi-algorithm approach is both successful and necessary in indicating such regions and appropriate learning algorithm selection is more dependent on image cluster interpretability rather than cluster validation indices. However, the Silhouette Width validation index consistently picks the algorithm providing the result with most interpretability. Regions to be subjected to river sediment deposition were confirmed when validated against satellite images of the same region captured in the future. Such techniques can be employed in policymaking, disaster management, labeling for supervised learning etc.

1. Introduction

The nature of human knowledge as a double edged sword has become very apparent in the 21st century. The rise of the Anthropocene, ushering the invention of the internal combustion engine and the age of industrialization, has directly contributed to climate change and global temperature rise, instigating numerous global environmental issues, such as a weaker hydrologic cycle (Ramanathan et al., 2001). Implications and applications of technological knowledge has led to the eventual collision of nature and man in the form of natural events such as, flooding, sea level rise, saltwater intrusion and river sediment deposition as well. On the other hand, research into renewable and less polluting forms of energy can be considered the second edge to mankind's knowledge base. Our endeavor in this study is deeply rooted to the application of technological knowledge such as remote sensing and data science in generating insightful knowledge about the Earth's natural processes.

River sediment deposition, a term with several synonymic nomenclature, refers to the process by which physical matter transported by the riverine systems of the Earth's surface accumulates at specific geographic regions harboring certain hydrogeochemical and anthropogenic characteristics (Schumm, 1977; Yang, 1996; Morris and Jiahua, 1997). The hydrologic cycle not only has a profound impact on the global climate (Chahine, 1992; Ramanathan et al., 2001), it is the carrier of the physical sources of sediment deposition as well. As water moves from higher to lower altitudes, potential energy is transformed to kinetic energy which induces the movement of sediment, leading to river deposition, the sources and determinants of which, in some cases anthropogenic (Kuhnle et al., 1996) and in other cases natural (Wolman, 1967; Smith, 1976; Nahlik and Mitsch, 2008). The causes of river sediment deposition have been studied globally, e.g., in France (Steiger and Gurnell, 2003), China (Chien et al., 1980; Milliman et al., 1987; Xu, 1998, 2002), South America (Stallard, 1987), Australia (Gibling et al., 1998) etc. We are also aware about the effects of river sediment deposition that have been studied in Canada (Nanson and Beach, 1977), England (Brookes, 1986), Greece (Andredaki et al., 2014), Netherlands (Geerling et al., 2008), Europe (Humborg et al., 1997), China (Ren and Shi, 1986; Wright et al., 1990; Yang, 1999; Jiongxin, 2004), Bangladesh (Kuehl et al., 1989) etc. The impacts of sediment deposition have been studied in the context of the global oceans as well (Gibbs, 1981; Milliman and Meade, 1983).

Thus, it is perceptible that research on measurement and recording of attributes of river sediment and deposition through various perspectives is both necessary and extant (Folk and Ward, 1957; Lambert

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and Walling, 1987; Ritchie and McHenry, 1990; Walling and He, 1997; Lane et al., 2003; Baborowski et al., 2007; Walling, 2017). Studies on predictive modeling of river sediment deposition have also been performed over the years (Benda and Cundy, 1990; Nicholas and Walling, 1997; Mulder et al., 1998; Scully et al., 2002; Lynds and Hajek, 2006).

In recent times, data driven machine learning techniques have been employed in modeling and prediction of sedimentation (Bhattacharya and Solomatine, 2006; Azamathulla et al., 2010). However, approach of applying unsupervised learning (Hastie et al., 2009) on remotely sensed satellite images for sediment deposition forecasting does not exist. Contrary to supervised learning, where the target variable is available and defined, the term 'unsupervised' in unsupervised learning refers to the fact that the target variables/classes are not available/provided to the learning algorithms. This translates to the suitability of unsupervised learning techniques for the purpose of this study due to unavailability of ground truth data (actual classes/target classes). Several unsupervised learning techniques exist which have been applied in various fields. For example, k-means, hierarchical agglomerative clustering (HAC), self-organizing tree algorithm (SOTA) and Clara learning techniques have been applied in medical image analysis (Pham et al., 2000; Zheng et al., 2003; Lasch et al., 2004; Shi et al., 2005; Ng et al., 2006; Li et al., 2011), for biomolecular sequencing (Fraga et al., 1998; Luo et al., 2004) in web data mining (O'Connor et al., 2010; Johnson and Gupta, 2012) etc. The wide use and success of the aforementioned unsupervised learning algorithms in multiple fields instigates this study to employ them in the context of remote sensing. Unsupervised learning techniques leverage the power of separating datum based on the characteristics of observations of the sample/population, when applied to satellite images, can be used for obtaining regions most susceptible to river sediment deposition. In a riverine country such as Bangladesh, where the rivers have significantly distinct characteristics compared to other regions across the globe, it is a suitable location for the application of aforementioned techniques for forecasting river sediment deposition.

Bangladesh is a very densely populated country with a massive proportion of the population relying on the fertile delta. Transfer of fertile sediment to the delta areas, naturally fertilizes land, which is integral for the agro-economy of the country. The Brahmaputra and Jamuna rivers of Bangladesh have been studied in the context of historical river channel migration and metamorphosis (Coleman, 1969; Bristow, 1987; Bristow et al., 1999), geomorphological structure (Ashworth et al., 2000; Richardson and Thorne, 2001; Best et al., 2003, 2007), sediment mineralogy (Datta and Subramanian, 1997), current flow dynamics (Richardson and Thorne, 1998; Klaassen, 1999) etc. Assessment of sediment of Karnaphuli River (Ali et al., 2016) and Meghna River (Anawar et al., 2002) have also been studied. Aforementioned studies are integrally anastomosed to river sediment deposition and these works have emphasized on physical inducers and characteristics of river sediment and its deposition process as well as its hydromorphological effects. However, we are largely unaware of the results of applying proposed technique of forecasting river sediment deposition through unsupervised learning. Furthermore, no existing study utilizes machine learning based remote sensing applications in the context of forecasting river sediment deposition. In order to address this research gap, we have encountered several important questions – i) what are the determinant features (satellite image bands) affecting unsupervised learning? ii) what is the efficacy of application of multiple learning algorithms? iii) what is the selection criteria of these algorithms? and iv) is there any cluster validation metric which governs the usability of these algorithms? This study, first of its kind, aims to address these questions in the process of forecasting river sediment deposition by applying unsupervised machine learning techniques on remotely sensed satellite imagery.

2. Materials and methods

2.1. Data

The Landsat Mission is co-managed by United States Geological Survey (USGS) and National Aeronautics and Space Administration (NASA) providing georeferenced satellite images for research purposes. Since the study spans 30 years (1988–2018), images of 1988, 1998, 2008 and 2018 (USGS, 2018) have been collected. Images of Kazipur, Sirajganj in Bangladesh (between latitudes 24°40′30.0″N and 24°37′30.0″N and between longitudes 89°37′12.0″E and 89°43′12.0″E), a region of the country being the site of continuous river erosion and sedimentation, primarily due to close proximity to the Jamuna River (Hossain and Sakai, 2008), makes it a suitable region for applying remote sensing based river sediment deposition forecasting techniques. The satellite images of the region were collected through Google Earth Engine (Gorelick et al., 2017) as TIF files. Due to intra-annual changes in dynamics of river flow, image collection seasons were kept constant (late winter) to avoid bias of such variability while comparing inter-annual results. Satellite imaging sensors capture information across wide ranges of the electromagnetic spectrum, not just the red, green and blue (RGB) spectrum. Bands capturing information in the non-visible infrared spectrum (NIR, SWIR, TIR) provides crucial information for predicting regions to be subject to deposition. Cloud cover of collected images were under 5% to avoid providing incorrect pixel values to unsupervised learning algorithms. The satellite image sensor bands being used as the features are shown in Table 1, and statistical summary of features of collected images are shown in Table 2. Therefore, individual pixels of images of 1988, 1998 and 2008 has 7 features while pixels of image of 2018 has 11 features. Scaling of images to spatial resolution of 90 m produces images of height 60 pixels and width 112 pixels, resulting in each image consisting of 6720 pixels i.e., training examples.

2.2. Preprocessing

The image preprocessing steps are discussed in details chronologically in this subsection. Landsat satellite images are available at a spatial resolution of 30 m on Google Earth Engine. Unsupervised learning is an iterative process, making it a computationally expensive process. The satellite images of scene with path 138 and row 43 are clipped according to rectangular study area shapefile containing the regions of different stages of river sediment deposition (for details see Section 3.5) which are then exported at 90 m resolution to reduce the training time since the total number of pixels are reduced by a factor of 9 i.e. 9 pixels at 30 m spatial resolution cover the same area as 1 pixel at 90 m spatial resolution. The images are downloaded as Tagged Image Format (TIF) files, then converted to comma separated value (CSV) files using the Geospatial Data Abstraction Library (GDAL) (GDAL/OGR Contributors, 2018) package of Python 3. Each CSV files contains the specific band values of a specific image resulting in 32 files (7 bands for 3 Landsat-5 images and 11 bands for 1 Landsat-8 image), which are then converted to 4 CSV files (one for each year's image) where each row corresponds to a single pixel and each column corresponds to the band values for the pixels. The resulting CSV files are standardized using the previously calculated mean and standard deviation of band values (Table 2). Standardization of data further reduces the training time required by the learning algorithms. The values x_{jb}, where j is the index of the training data and b is the band with mean μ_b and standard deviation σ_b, are standardized using the following formula

$$x_{jb} = \frac{x_{jb} - \mu_b}{\sigma_b}$$

Table 1
Features of satellite images of Landsat 5 and 8 used in this study.

Satellite	Sensor	Path/Row	Spatial Resolution (m)	Spectral Band number	Spectral Band Name	Wavelength (micrometers)
Landsat-5	Thematic Mapper (TM)	138/43	30	Band 1	Blue	0.45–0.52
				Band 2	Green	0.52–0.60
				Band 3	Red	0.63–0.69
				Band 4	Near Infrared (NIR)	0.76–0.90
				Band 5	Shortwave Infrared (SWIR) 1	1.55–1.75
				Band 6	Thermal	10.40–12.50
				Band 7	Shortwave Infrared (SWIR) 2	2.08–2.35
Landsat-8	Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS)	138/43	30	Band 1	Ultra Blue (coastal/ aerosol)	0.435–0.451
				Band 2	Blue	0.452–0.512
				Band 3	Green	0.533–0.590
				Band 4	Red	0.636–0.673
				Band 5	Near Infrared (NIR)	0.851–0.879
				Band 6	Shortwave Infrared (SWIR) 1	1.566–1.651
				Band 7	Shortwave Infrared (SWIR) 2	2.107–2.294
				Band 8	Panchromatic	0.503–0.676
				Band 9	Cirrus	1.363–1.384
				Band 10	Thermal Infrared (TIRS) 1	10.60–11.19
				Band 11	Thermal Infrared (TIRS) 2	11.50–12.51

Table 2
Statistical summary of pixel values of satellite images of 1988, 1998, 2008 and 2018.

Year	Band	Minimum	Maximum	Mean	Standard deviation
1988	Blue	0.1342	0.2234	0.1572	0.0172
	Green	0.1068	0.2348	0.1434	0.0223
	Red	0.0881	0.2421	0.1279	0.0294
	NIR	0.0665	0.3275	0.1728	0.0532
	SWIR 1	0.0123	0.3848	0.1530	0.0813
	Thermal	289.71	295.52	292.43	1.1459
	SWIR 2	0.0031	0.3711	0.1172	0.0803
1998	Blue	0.1162	0.2306	0.1519	0.0307
	Green	0.0972	0.2466	0.1448	0.0395
	Red	0.0759	0.2634	0.1337	0.0493
	NIR	0.0510	0.3088	0.1660	0.0641
	SWIR 1	0.0128	0.3697	0.1533	0.1028
	Thermal	291.53	303.97	296.46	3.4216
	SWIR 2	0.0098	0.3605	0.1276	0.1049
2008	Blue	0.1559	0.2102	0.1759	0.0114
	Green	0.1356	0.2125	0.1604	0.0162
	Red	0.1118	0.2115	0.1432	0.0220
	NIR	0.0877	0.2537	0.1620	0.0383
	SWIR 1	0.0253	0.3343	0.1456	0.0878
	Thermal	288.79	297.26	292.46	2.0478
	SWIR 2	0.009	0.3251	0.1137	0.0885
2018	Ultra Blue	0.1469	0.2283	0.1660	0.0175
	Blue	0.1221	0.2232	0.1463	0.0219
	Green	0.0946	0.2277	0.1282	0.0283
	Red	0.0716	0.2437	0.1147	0.0375
	NIR	0.0482	0.4043	0.1619	0.0671
	SWIR 1	0.0158	0.3626	0.1314	0.0959
	SWIR 2	0.0089	0.3534	0.1040	0.0969
	PAN	0.0859	0.2308	0.1223	0.0311
	Cirrus	0.0002	0.0015	0.0008	0.0001
	TIRS 1	291.13	303.80	295.88	3.1058
	TIRS 2	289.96	301.81	294.30	2.9377

2.3. Pearson-product moment correlation coefficients

The underlying concept of unsupervised learning is based on the presumption that observations are separated based on how their feature values change/differ. However, it is possible for many features to have similar characteristics, dissimilar characteristics or no semantic dynamic relationship. It is necessary to explore the dynamics of the interconnected relationship among features because the interpretation of results of unsupervised learning may depend on the interpretations of such relationships among the features (band values) of the observations (pixels) themselves. The Pearson-product moment correlation coefficient (Pearson, 1907) is used for analyzing the relationship between the features. The coefficient of each of the feature pairs are calculated, analyzed and compared to reveal the semantic relationship between the features.

2.4. Algorithms

Prediction of river deposition via supervised classification of satellite image pixels involves manually labeling specific regions for use as training and testing examples, which is not possible if cloud free and high quality satellite images of the near future are not available. Thus, unsupervised learning techniques are employed for the purpose of the study.

2.4.1. k-means

The k-means algorithm (Hartigan and Wong, 1979) divides n vectors into k clusters where each vector is a member of the cluster with the closest mean value. The aim is to minimize the squared error function J given by,

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} (x_i^{(j)} - \mu_j)^2$$

where $x_i^{(j)}$ is a member of the jth cluster, indexed as the ith vector and μ_j is the mean of the jth cluster.

2.4.2. Hierarchical agglomerative clustering (HAC)

The complete linkage method (Wilks, 2011) also known as the farthest neighbor clustering, is used. The algorithm starts by placing each vector in its own cluster then merges the clusters repetitively. The complete linkage method considers the distance D given by,

$$D(X, Y) = \max_{x \in X, y \in Y} d(x, y)$$

between two clusters X and Y, to be the farthest possible distance by choosing a member x from the first cluster and a member y from the second cluster such that the distance d between the vectors of the two

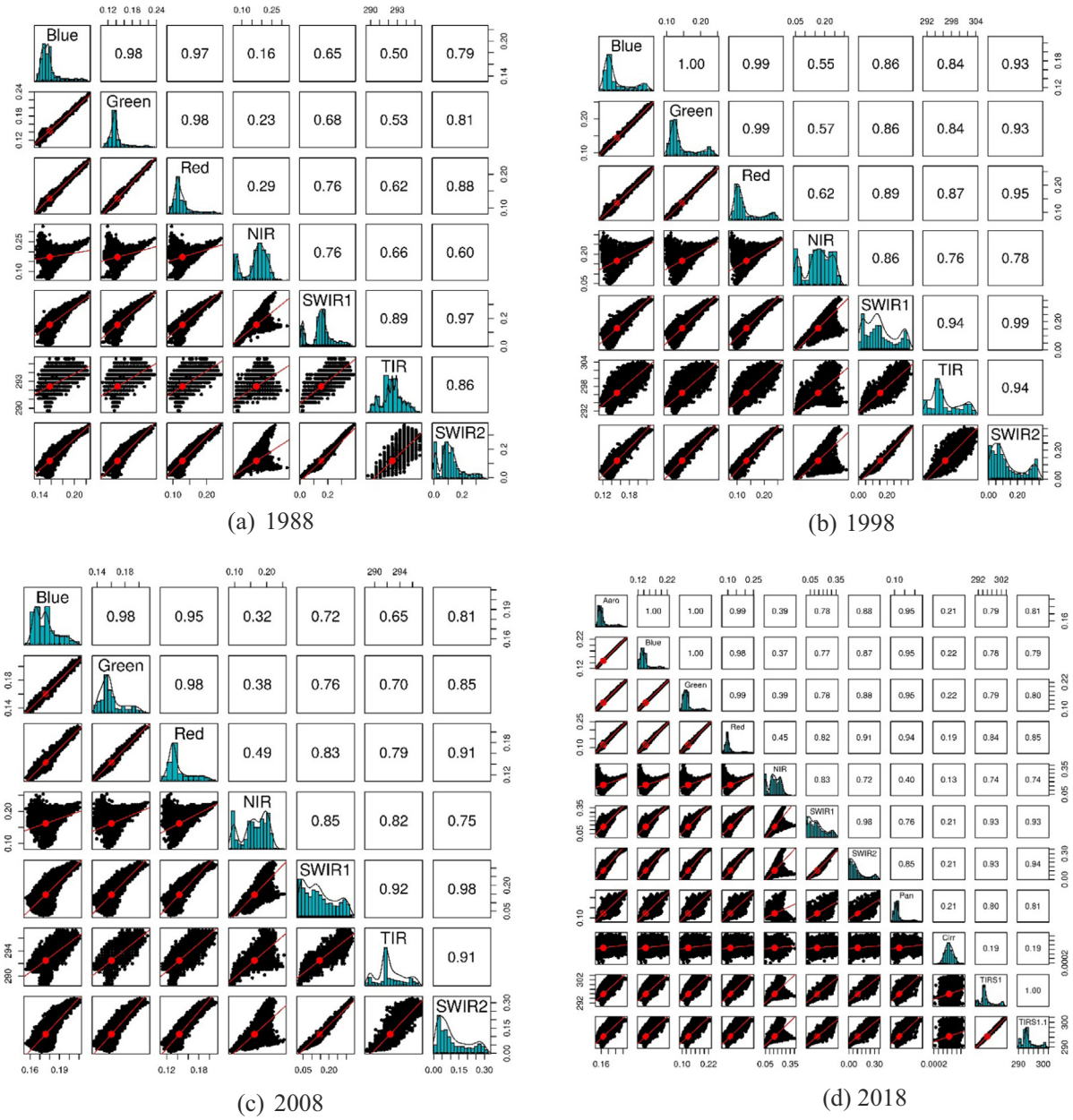


Fig. 1. Pearson-product moment correlation among features of images of (a) 1988, (b) 1998, (c) 2008 and (d) 2018.

clusters is maximized.

2.4.3. Self-organizing tree algorithm (SOTA)

SOTA (Herrero et al., 2001) is a combination of Self-Organizing Maps (SOM) (Kohonen, 1990) and hierarchical clustering. Selected nodes (clusters) with highest variance between members are split into two clusters repetitively. At the end of splitting, each leaf represents a cluster. The Euclidean distance metric is used.

2.4.4. Clustering large applications (Clara)

The algorithm divides entire data set of vectors into smaller samples to reduce complexity and uses Partitioning Around Medoids (PAM) (Kaufman and Rousseeuw, 2009) on samples. The aim is to minimize the cost function J given by,

$$J(M, N) = \frac{\sum_{i=1}^{n} h(x_i, r(M, x_i))}{n}$$

where M is a set of selected medoids, N is the dataset of n vectors, $h(x_i, x_j)$ is the heterogeneity between vectors x_i and x_j, and $r(M,x_i)$ returns the medoid in M closest to x_i. The Euclidean distance metric is used.

2.5. Optimum cluster number selection

In unsupervised learning, selecting the number of clusters/classes is integral to achieving target results. The same number of clusters have to be provided to all four algorithms to ensure that results of labeling a specific satellite image vary from one algorithm to another solely based on how well the algorithms are able to cluster the pixels, rather than being affected by variable number of clusters provided to each of the algorithms. In most unsupervised learning algorithm software packages, it is a requirement to provide the number of clusters before performing computation, adding to the need of determining the optimum number of clusters. Selecting a higher than optimum number leads to too many clusters, where the target cluster may get divided into two or more clusters reducing interpretability. Whereas selecting a

Table 3
Number of clusters and corresponding number of supporting metrics.

Number of clusters	Number supporting metrics
2	2
3	1
4	13
5	0
6	1
7	1
8	1
9	0
10	2

lower than optimum number of cluster may cause the target cluster to be included as part of a larger cluster, where its cluster members are very dissimilar i.e., band values of member pixels differ by a large margin. The optimum number of clusters are obtained by comparing 21 different validation metrics. The number of cluster(s) with highest number of supporting validations metrics is chosen.

2.6. Cluster validation

Numerical measures of the performance of unsupervised learning algorithms (cluster validation metrics) depend on how well an algorithm is able to form clusters i.e., how similar the members of the same clusters are. Such metrics are numerous and allows to both assess the performance of an unsupervised learning algorithm as well as compare performances between two or more algorithms. Of the 21 such metrics used for optimum cluster number selection, three have been employed again for the task of cluster validation viz., the Connectivity Measure, the Dunn index and the Silhouette Width. The Connectivity Measure (Handl et al., 2005) takes values between 0 and ∞, where a higher value is preferred i.e. a value close to 0 refers to poor clusters and close to ∞ refer to almost perfectly clustered pixels. The Dunn Index (Dunn, 1974) has values between 0 and ∞ and the aim is to maximize the Dunn Index. The Silhouette Width (Rousseeuw, 1987) falls within a range between −1 and 1, where values close to −1 denote poorly clustered data and values close to 1 denote well clustered data. The cluster validation index values are a numerical measure of how similar the pixels of the same clusters are by calculating the distance (in band feature space) between the pixels. It translates into the ability of an algorithm in forming clusters with member pixels having approximately equal band/feature values.

2.7. Software uses

The GDAL Library (GDAL/OGR contributors, 2018) of Python 3 is used for converting satellite image TIF files to CSV files. The R Language (R Core Team, 2013) is used for employing four aforementioned unsupervised learning algorithms. Nbclust package (Charrad et al., 2014), psych package (Revelle, 2018) and clValid package (Brock et al., 2008) have been used for finding optimum number of clusters, for pairwise plotting of feature bands and for cluster validation respectively.

3. Results and discussion

3.1. Pearson-product moment correlation coefficient analysis

The correlation coefficient of the satellite image bands (features) is shown in Fig. 1. Each feature has been compared against every other feature to explore the relationship between how the features change values with respect to each other. The Red, Green and Blue (RGB) bands are highly correlated (positively) to each other. High correlation of RGB bands is evident not only in images of individual years, but in images of all years, thus implying that employing just the Red, Green and Blue bands as features is not sufficient for properly separating pixels since the three bands are almost perfectly correlated. On the other hand, the NIR, SWIR (1 and 2) and TIR (1 and 2 for Landsat-8) (NST) bands show remarkable differences in correlation to the RGB bands. The NST bands also differ in the magnitude of the correlation to RGB band in images of different years. The SWIR and TIR bands are more correlated to RGB bands than the NIR band, signifying the importance of the NIR band as a feature for unsupervised learning. The panchromatic band of Landsat-8 are highly correlated to RGB bands, implying issue of being a non-determinant band for unsupervised learning. The cirrus band (used for cloud detection) of Landsat-8 is not highly correlated to any other band (feature). The reason for the RGB bands having similar characteristics across images of all years is due to similar land cover composition in the visual spectrum. Whereas the NST bands can be considered the determinant features since these are the bands that actually vary and are most significant in determining the resultant cluster of an individual pixel.

3.2. Optimum cluster number

Image of 1988 has been used as benchmark for finding the optimal number of clusters, since land cover composition ratio i.e. land, water, newly accreted land etc. are similar across images of all years. The range to be tested for finding optimum number of clusters is between 2 and 10 inclusive, since more than 10 clusters, even with well validation score, might be too uninterpretable. The process involves clustering image of each year into clusters (2–10 inclusive) and calculating the number of validation metrics that 'choose' a specific number of cluster as the most optimal and then choosing number of cluster with most metrics supporting the choice of that number. Results portrayed in Table 3 indicate that 13 of 21 validation metrics show the optimum number of clusters is 4 for the purpose of the study.

3.3. Cluster validation

The validation scores in Tables 4 and 5 show that HAC is a consistently well performing algorithm, having the best score for both Connectivity Measure and Dunn Index for all years. Whereas SOTA fails to achieve any optimal validation score. k-means algorithm has optimal Silhouette Width for satellite images of years 1998 and 2008. From mathematical and numerical perspective of validation scores, which depends on how similar the members of a cluster are, it is evident that HAC algorithm is performing the best if each of the three cluster validation metrics (Connectivity, Dunn Index and Silhouette Width) are considered to have equal ability in choosing of algorithm which provides resultant clustered images with highest interpretability.

Table 4
Cluster validation measures of learning algorithms applied on satellite images.

Year	Algorithm	Connectivity	Dunn index	Silhouette width
1988	k-means	393.19	0.0099	0.4847
	HAC	159.03	0.0341	0.5165
	SOTA	506.90	0.0069	0.2639
	Clara	428.59	0.0081	0.4575
1998	k-means	359.28	0.0063	0.5102
	HAC	251.43	0.0304	0.4908
	SOTA	438.79	0.0084	0.4603
	Clara	323.87	0.0229	0.5100
2008	k-means	345.62	0.0206	0.4871
	HAC	243.99	0.0350	0.4194
	SOTA	619.23	0.0112	0.4215
	Clara	369.41	0.0158	0.4870
2018	k-means	686.92	0.0096	0.3619
	HAC	422.15	0.0342	0.3316
	SOTA	783.30	0.0188	0.3101
	Clara	606.13	0.0214	0.3630

Table 5
Algorithm with optimal scores of each cluster validation measures.

Year	Validation measure	Algorithm
1988	Connectivity	HAC
	Dunn Index	HAC
	Silhouette Width	HAC
1998	Connectivity	HAC
	Dunn Index	HAC
	Silhouette Width	k-Means
2008	Connectivity	HAC
	Dunn Index	HAC
	Silhouette Width	k-Means
2018	Connectivity	HAC
	Dunn Index	HAC
	Silhouette Width	Clara

3.4. Image cluster analysis

For the year 1988, HAC algorithm has optimal scores for all three cluster validation indices (Connectivity, Dunn index and Silhouette Width). Even though Connectivity and Dunn Index are more optimal for HAC being used to cluster images of all years (1988, 1998, 2008 and 2018), results obtained by algorithm with optimum Silhouette Width scores are more interpretable (Fig. 2). The HAC algorithm applied on images of all years, consistently achieves optimal scores for Connectivity and Dunn Index. However, photointerpretation reveals that resultant clusters are highly error prone for the years 1998, 2008 and 2018 using HAC. Thus, for the purpose of the study, algorithm with optimal Silhouette width scores are considered to be the optimal unsupervised learning technique leading to selection of HAC for 1988, k-means for the years of 1998 and 2008, and Clara for 2018. Therefore, it is evident that the interpretability, and thus usability, of a specific algorithm is in fact not determined uniformly by all cluster validation metrics. Rather, the algorithm with the best Silhouette Width score has the highest interpretability, owing the usability in forecasting possible future river sedimentation regions. Fig. 3 shows the original RGB satellite images and corresponding clustered images using algorithm with optimal Silhouette Width score.

3.5. Deposition forecast interpretation

This subsection describes and discretizes the potential stages of river sediment deposition in the context of this study (Fig. 4), summarizes the chronological deposition stage classes (based on Figs. 2, 3 and 4) and their corresponding clusters in resultant images (Table 6), and finally provides the validation and interpretation of the forecasting of river sediment deposition (Fig. 5).

River sediment deposition is a continuous process which is affected by numerous variables. However, for the analysis and interpretation of the results of this study, it is necessary to discretize the stages of river sediment deposition. The height h_i of a specific region relative to the water surface level determines the stage it is in i.e., a region in Stage i has height h_i. Five stages of river sediment deposition have been considered viz., i) *Stage 0* regions correspond to the permanent river bed ii) *Stage 1* regions are part of the river bed being subjected to deposition and is the primal focus of this study, iii) *Stage 2* refers to regions which have been subjected to sufficient deposition to be approximately at the same level as the river water surface i.e. $h_2 \approx 0$, iv) *Stage 3* regions are the newly accreted land (the successor state of regions in Stages 0–2) and v) *Stage 4* is the permanent terrestrial regions. Considering the water surface level as the reference height and the downwards direction to be the positive direction, h_0 and h_1 have positive values due to being under the water surface level whereas h_3 and h_4 have negative values due to being above the water surface level, shown mathematically by

$$h_0 > h_1 > h_2 > h_3 > h_4$$

Analysis of image of 1988 (Fig. 5a and d) and corresponding resultant clusters (Fig. 5b and e) are validated by comparing to image of the same region taken 32 days later (Fig. 5c and f). Further inspection shows interesting characteristics (both similarities and differences) of the four clusters. It is evident from Table 6, Fig. 5b and e that regions which are to be subject to deposition i.e. Stage 1 (Cluster 2) regions are indistinguishable from Stage 0 (Cluster 1) regions if only RGB images are interpreted (Fig. 5a and d). However, these regions are being clustered separately from the river areas due to satellite images capturing information across a wide range of the electromagnetic spectrum

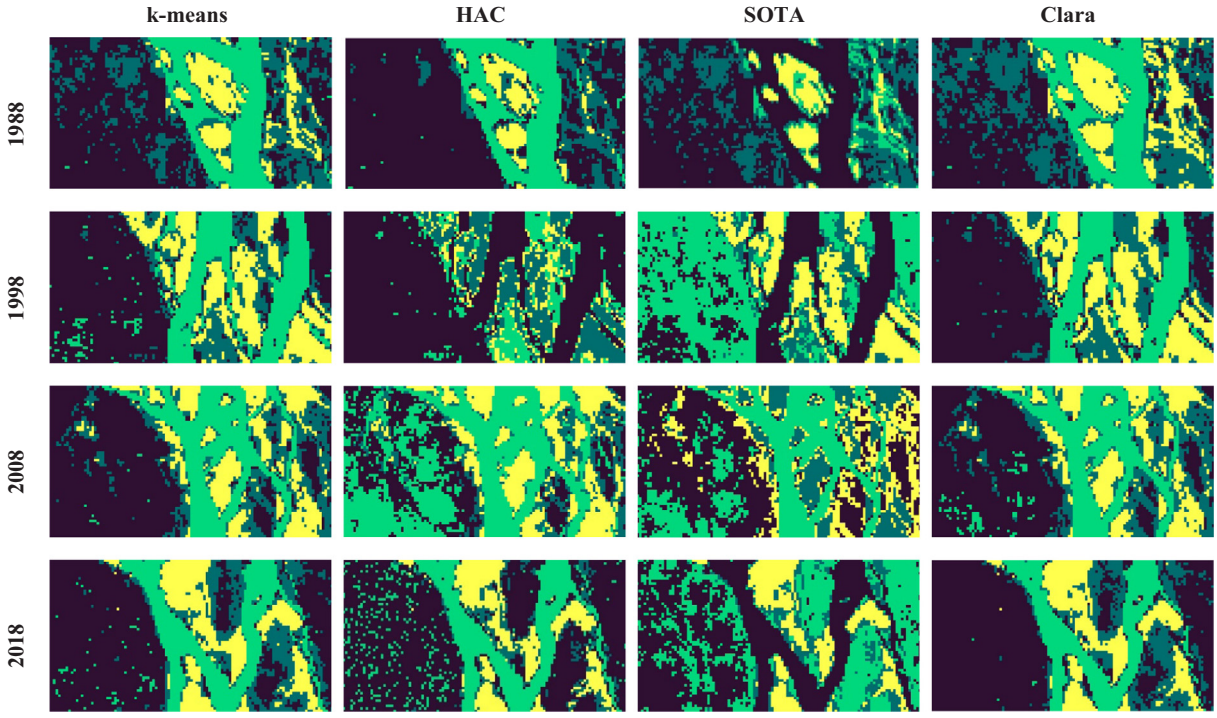


Fig. 2. Images of 1988, 1998, 2008 and 2018 clustered using all four unsupervised learning algorithms viz., k-means, HAC, SOTA and Clara.

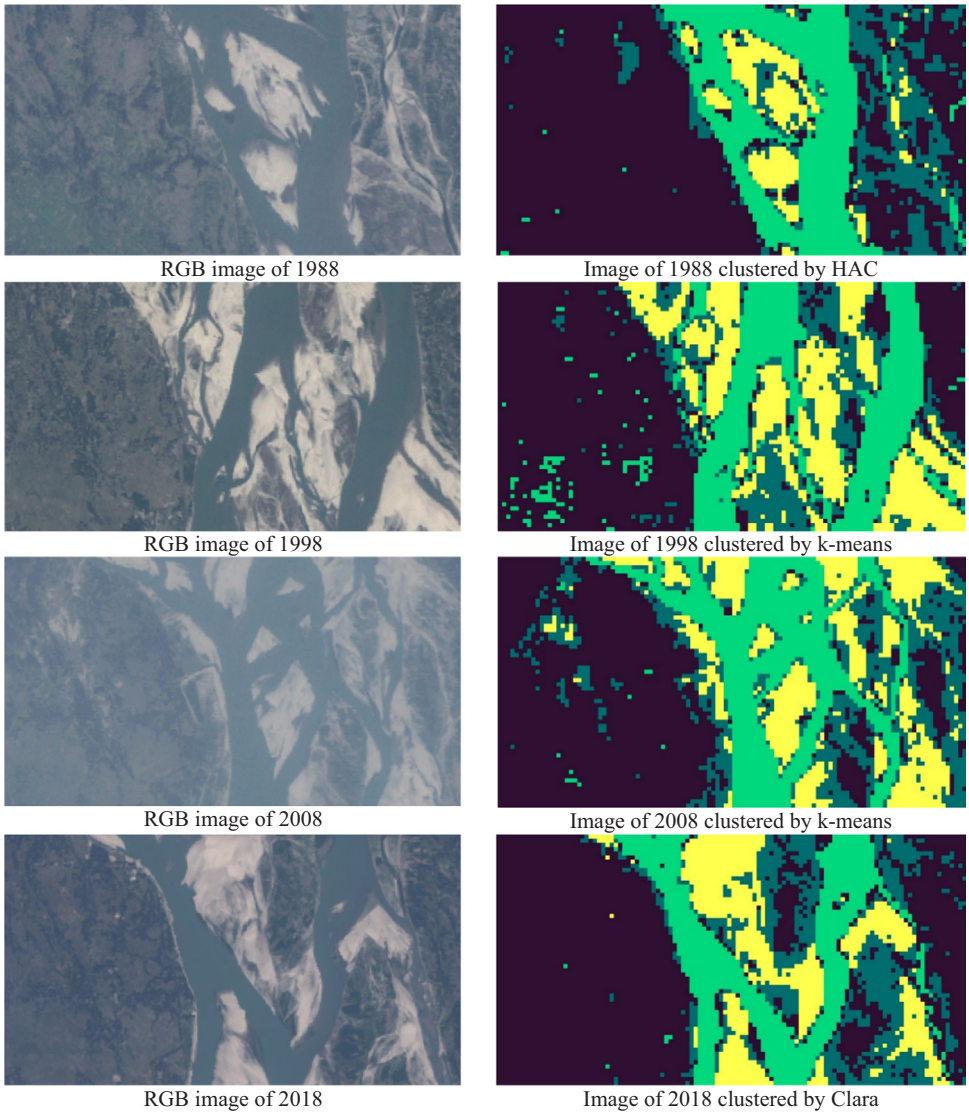


Fig. 3. RGB images and corresponding images clustered by algorithm with optimal Silhouette Width score.

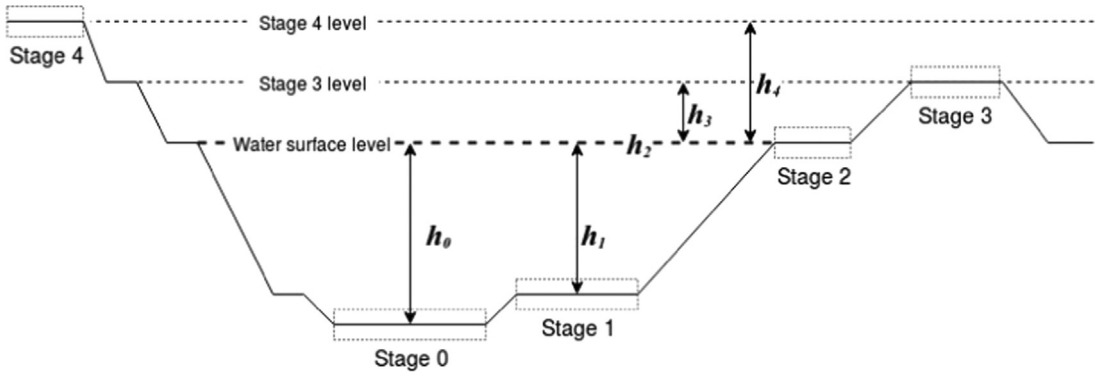


Fig. 4. Potential stages of river sediment deposition considered for this study.

and not just the RGB spectrum. Regions being subjected to river sediment deposition, which are underwater, gradually accretes sediment over time, decreasing the depth of the water over those regions. Since the primary physical difference between the river areas (Stage 0) and the regions susceptible to sediment deposition (Stage 1) is the water depth i.e. river areas have much greater depth than regions being subject to deposition, they also reflect radiation in different regions of the electromagnetic spectrum (which are recorded by the imaging sensors aboard the satellites).

It is also interesting to note that clustering algorithms do not include areas to be deposited (Stage 1, Cluster 2) with the clusters (Cluster 3 and 4) for existing deposited areas (Stage 2 and 3), rather most deeply submerged regions being subjected to deposition (Stage 1) are clustered with permanent land cover (terrestrial regions) (Stage 4), suggesting

Table 6
Deposition stage class and corresponding image clusters.

Cluster	Assigned color	Deposition stage class	Description of class
1		Stage 0	Regions of permanent river bed
2		Stage 1	Regions of river bed being subjected to deposition
		Stage 4	Permanent terrestrial regions
3		Stage 2	Regions at approximately the same level as the river water surface
4		Stage 3	Newly accreted land (the successor state of regions in Stages 0–2

that permanent land and depositional areas share non RGB features as electromagnetic radiation characteristics, possibly in the NST feature spectrum.

Thus it can be deduced that algorithm with optimal Silhouette Width score (HAC for 1988) is successful in clustering the different stages of sedimentation and is able to separate visually similar regions (with similar RGB features), confirmed by comparing with images captured 32 days later. The clustering of Stage 1 regions (which is the primal objective of the study) separately form Stages 0, 2 and 3, forecasts the locations of the regions which will eventually transits to Stages 2 and 3. The clustering results and validation image also confirms the transition of Stage 2 regions to Stage 3 regions. Even though images of such short span intervals (approximately 30 days) without cloud cover cannot be guaranteed, validation images do not affect clustered images and can be used for generating insightful spatial knowledge of river deposition regions.

4. Conclusion

Prediction of spatial regions susceptible to future river deposition via manual field based survey method is slow, error prone and expensive. Remote sensing, coupled with machine learning can be used to quickly predict such regions. The results indicate that unsupervised learning techniques are able to forecast depositional regions in rivers from satellite images and also shows that the interpretability provided by an unsupervised learning algorithm is governed by the Silhouette Width score. This study successfully shows that manual validation through images of same region collected in future date confirm results of forecasting. Unsupervised learning of satellite images can be vital tool to employ in disaster management and policymaking. Furthermore, unsupervised learning of satellite images to find clusters can be used as precursor to supervised satellite image pixel classification, given that clusters are validated by field-based ground truthing.

However, remote sensing based unsupervised learning approach is not free from limitations. Unsupervised learning techniques are able to forecast river deposition in the short term. Forecasting long term depositional regions would involve combining non-remotely sensed data such as total dissolved solid, river flow rate, river depth etc. with remotely sensed data. Furthermore quality of satellite images and availability along with cloud cover over region of interest determines the applicability of remote sensing based unsupervised learning techniques. Training time is an important factor governing the size of the study area considered for the study. Due to hardware limitations, only Kajipur, Sirajganj was considered as study area to reduce training time. Images clustered at 90 m resolution rather than 30 m resolution provide information at a reduced level of detail. Further research needs to consider more study areas and higher spatial resolution to propagate the applicability of this study.

Conflict of interest

None.

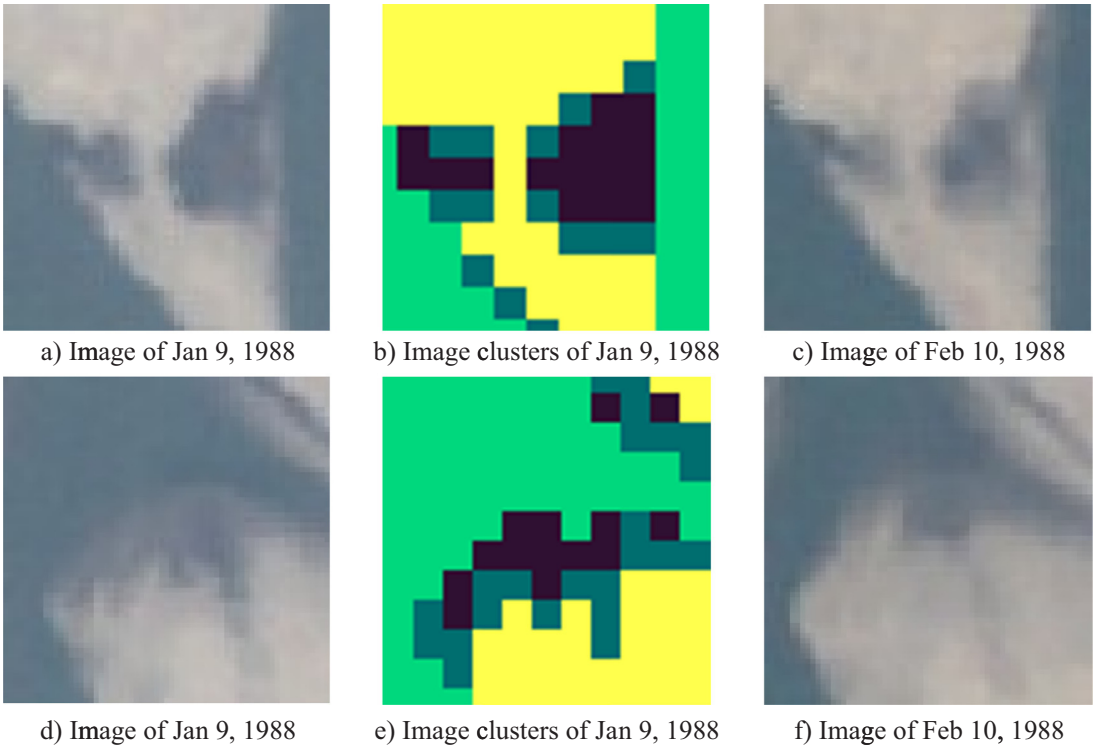


Fig. 5. River sediment deposition forecast validation.

Ethical statement

None.

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